



BobSim

ISO7401 Lateral Transient Response

Comprehensive Open-Loop Characterization

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- Step input (time-domain)
 - One-period sinusoidal input
 - Continuous sinusoidal response
 - Frequency response computed from fixed-amplitude, single-direction sine sweep

ISO7401 Metrics Summary

Time Domain — Step Response

STEP RESPONSE

Peak a_y	5.72	m/s ²
Steady-State a_y	5.35	m/s ²
Steady-State Gain	61.31	(m/s ²)/rad
Overshoot (a_y)	6.9	%
Rise Time (10–90%)	0.13	s
Settling Time	1.25	s
Yaw Overshoot	11.5	%

ISO7401 Metrics Summary

Frequency Domain — Sustained Sine

FREQUENCY RESPONSE — CORE

DC Gain (a_y/δ_H)	61.71	(m/s ²)/rad
DC Gain (r/δ_H)	3.10	(rad/s)/rad
Peak Gain (a_y)	61.71	(m/s ²)/rad
Peak Freq (a_y)	0.50	Hz
Bandwidth (-3 dB)	2.00	Hz
Phase @ 1 Hz (a_y)	-18.5	deg
Phase @ 1 Hz (r)	-12.0	deg
Lag @ 1 Hz (a_y)	0.051	s
Lag @ 1 Hz (r)	0.033	s

DYNAMIC CHARACTER

Gain Slope (a_y)	-1.05	dB/dec
Gain Slope (r)	0.54	dB/dec
Phase Slope (a_y)	53.04	deg/Hz
Phase Slope (r)	31.49	deg/Hz
Phase = -45° (a_y)	2.00	Hz
Phase = -45° (r)	2.00	Hz

RESPONSE COUPLING

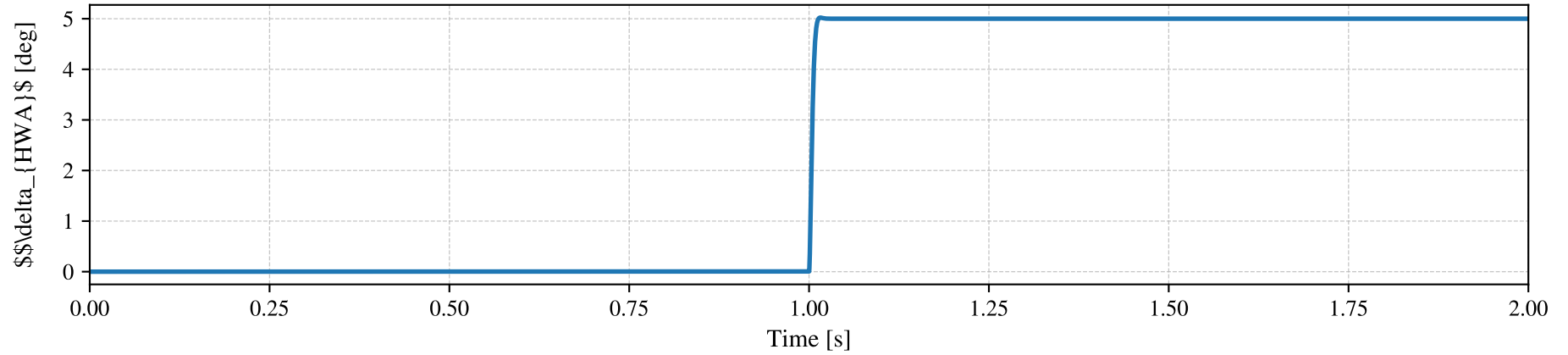
Lag: $\delta_H \rightarrow a_y$	0.051	s
Lag: $\delta_H \rightarrow r$	0.033	s
Lag: $r \rightarrow a_y$	0.018	s
r/a_y Ratio	0.051	(rad/s)/(m/s ²)

QUALITY / VALIDITY

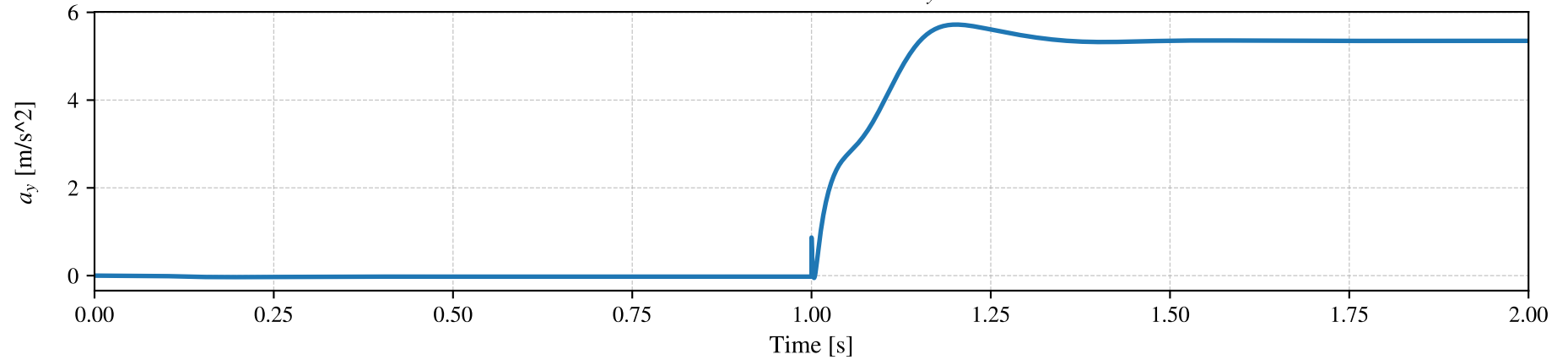
a_y Fit Error	1.01e-02	-
Yaw Fit Error	4.22e-03	-
Gain Variation	7.9	%

Step Input Response (ISO7401)

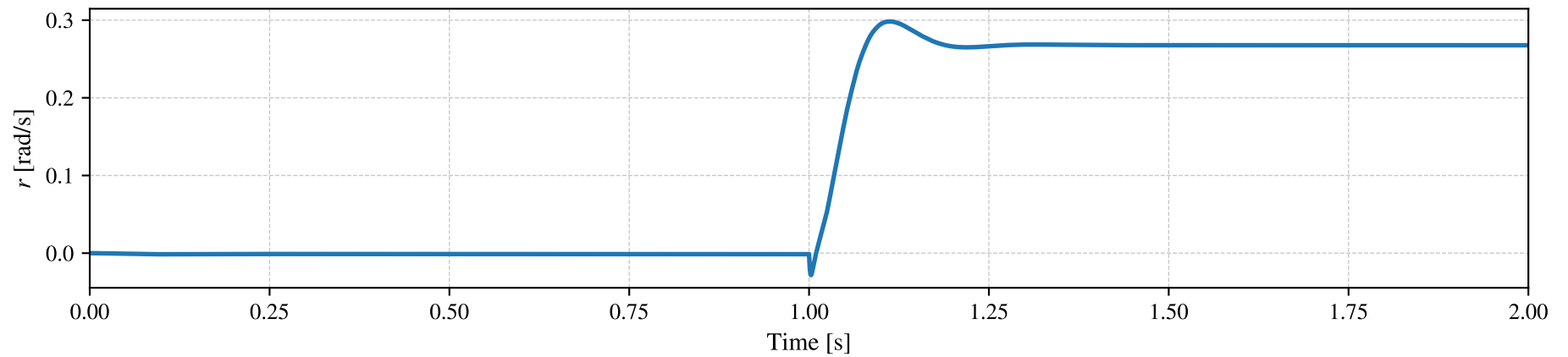
Steering Wheel Angle δ_H



Lateral Acceleration a_y

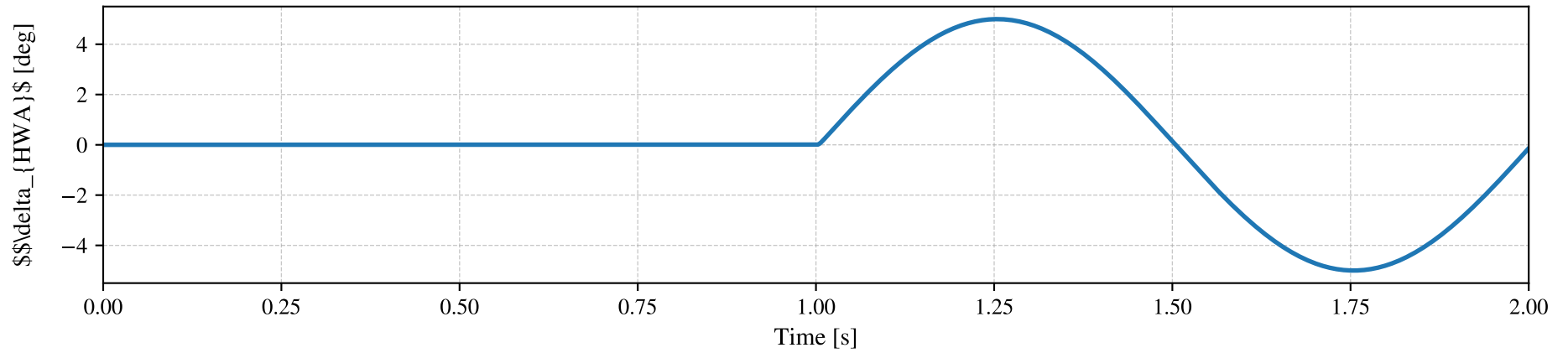


Yaw Velocity r

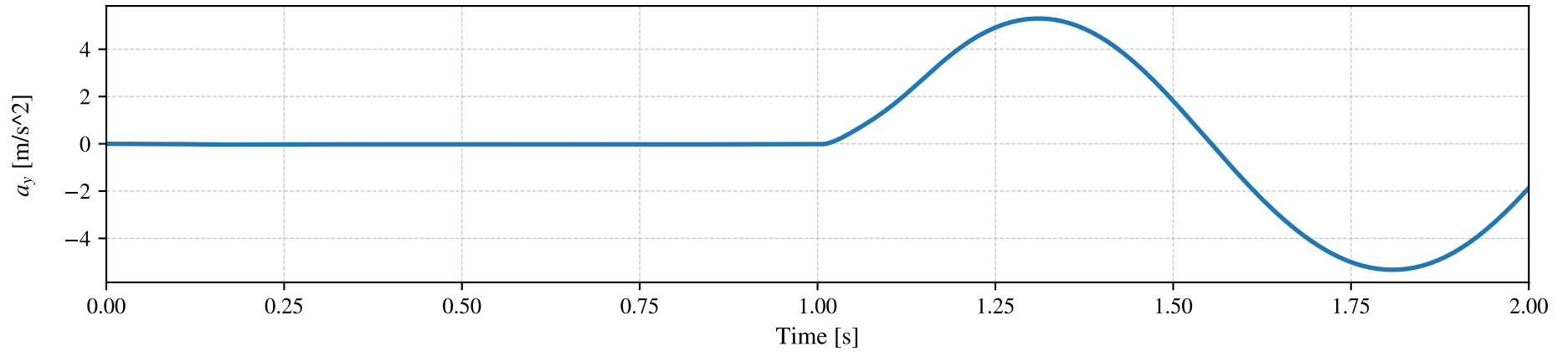


One-Period Sinusoidal Response (ISO7401)

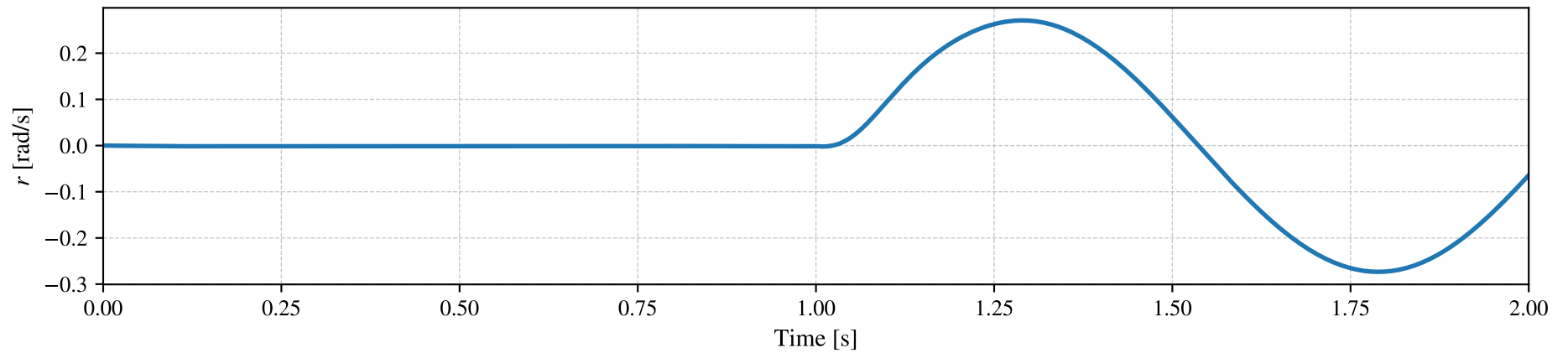
Steering Wheel Angle δ_H



Lateral Acceleration a_y

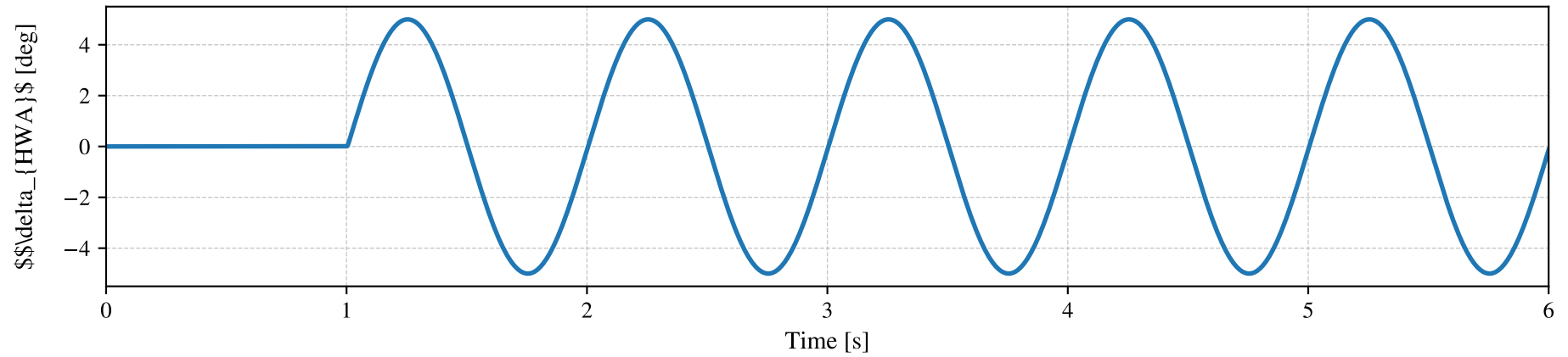


Yaw Velocity r

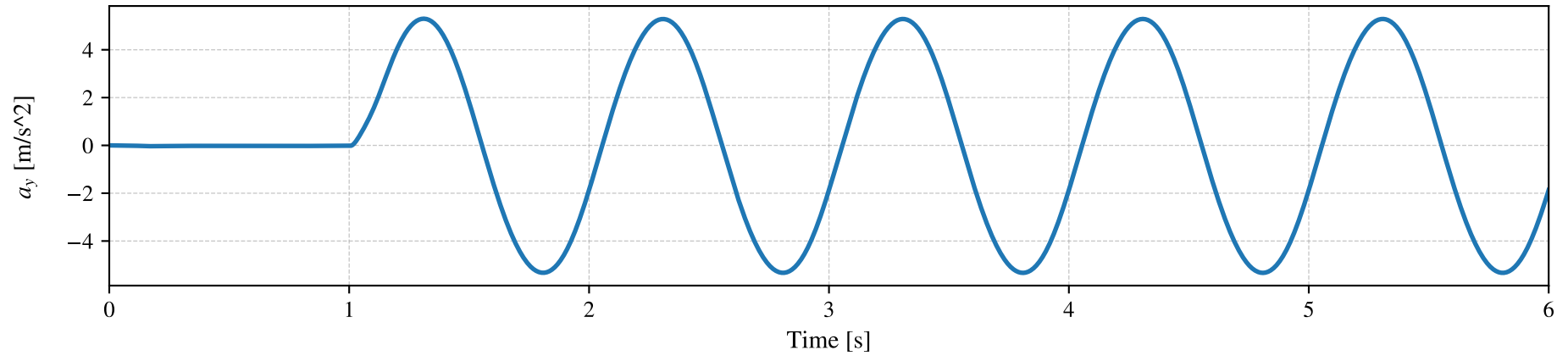


Continuous Sinusoidal Response

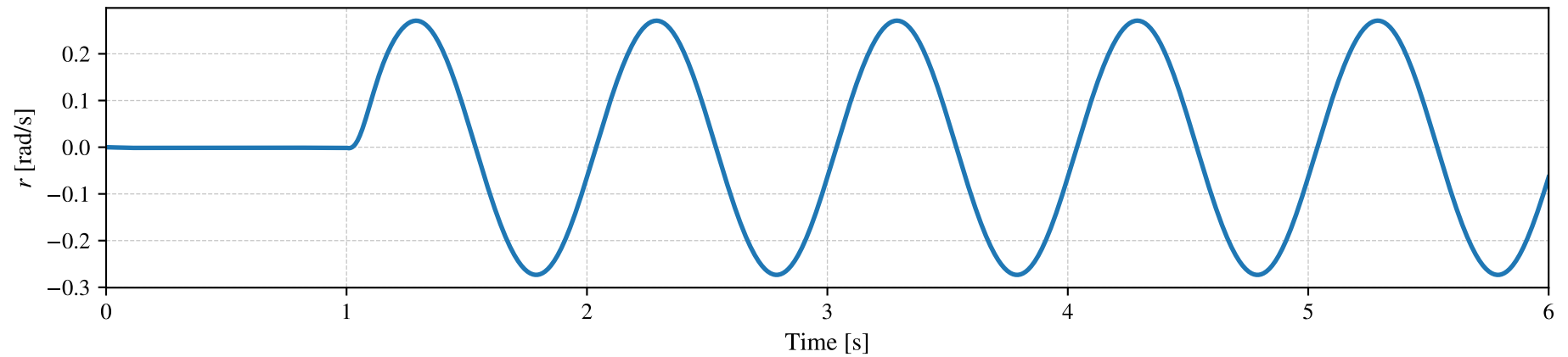
Steering Wheel Angle δ_H



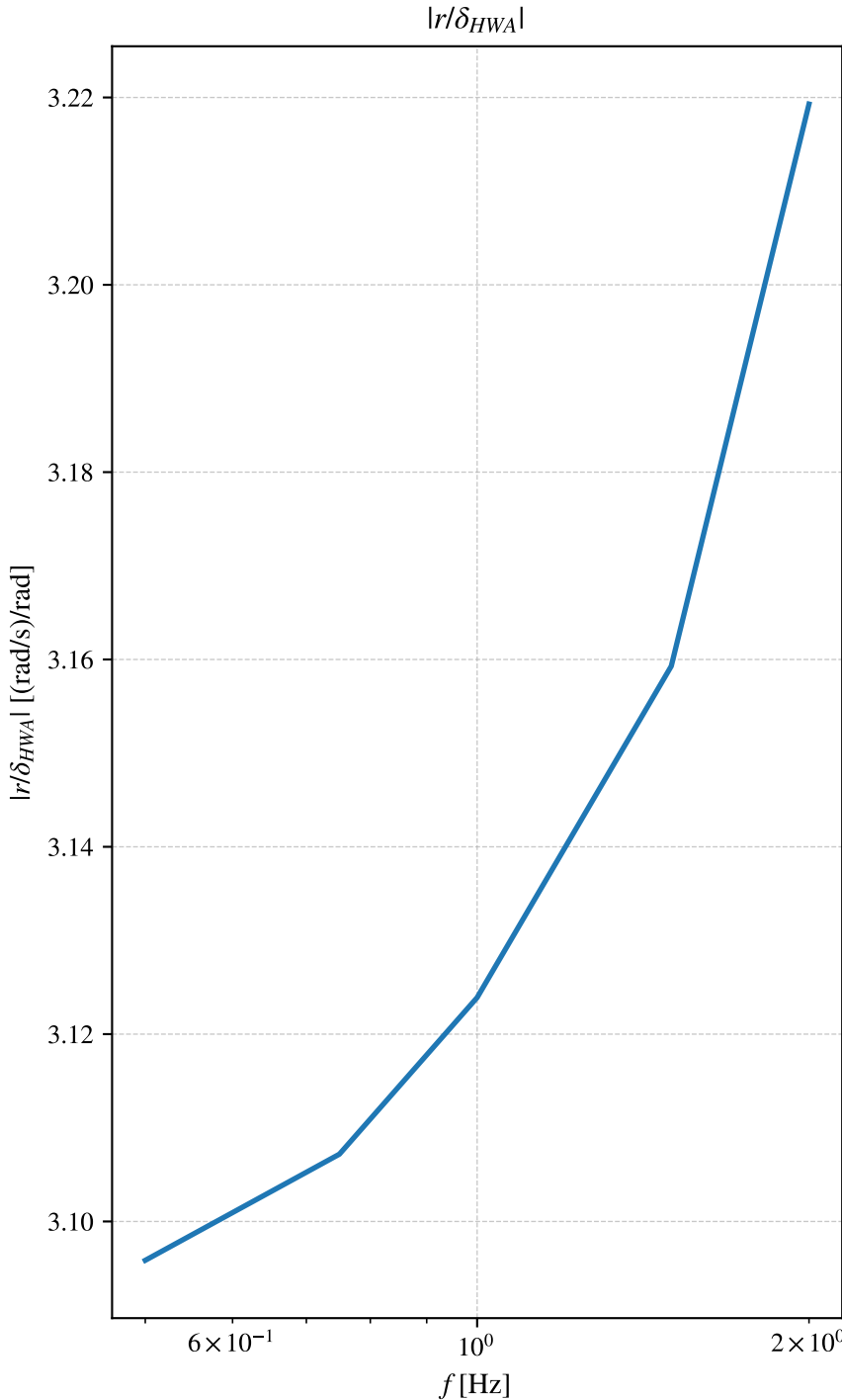
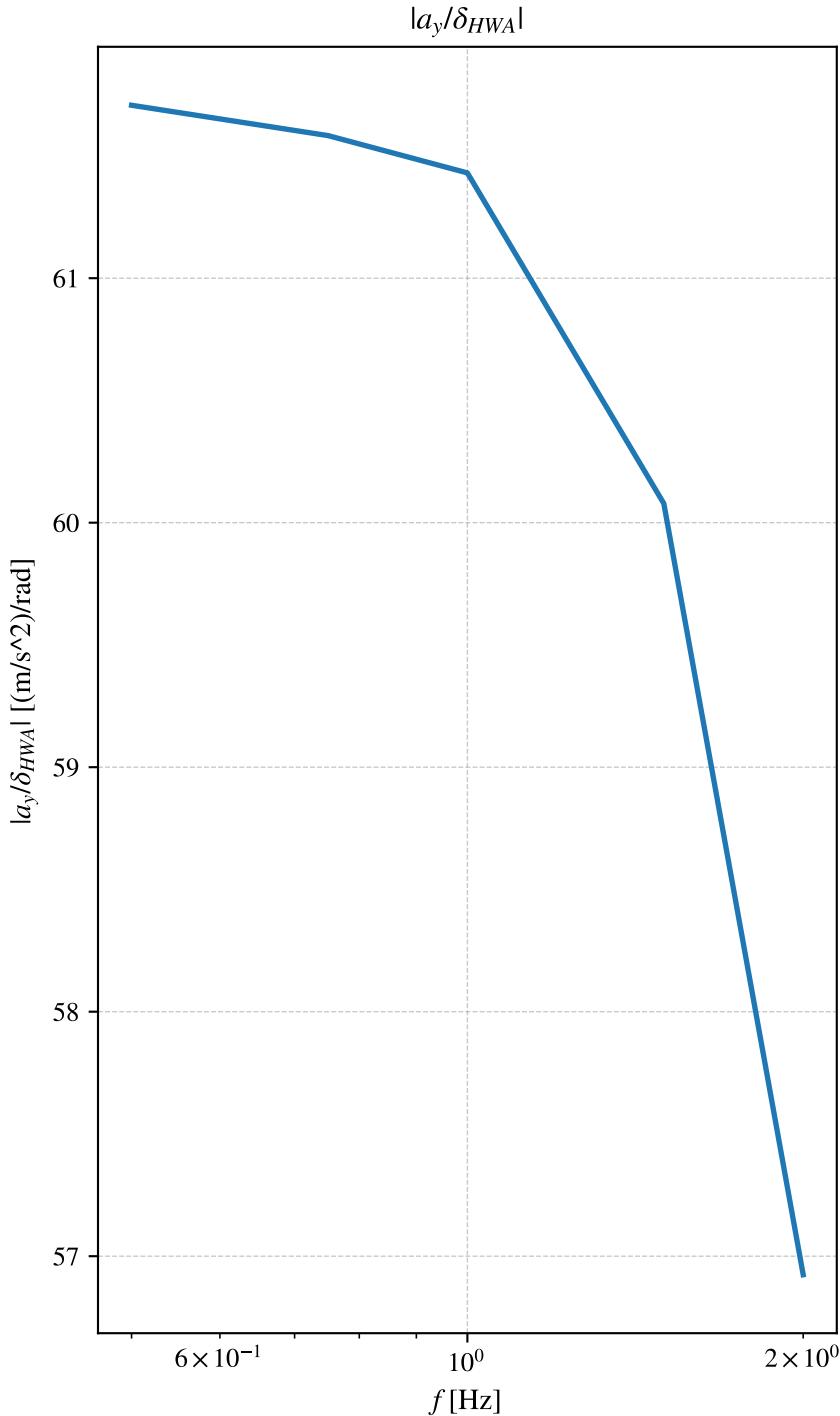
Lateral Acceleration a_y



Yaw Velocity r

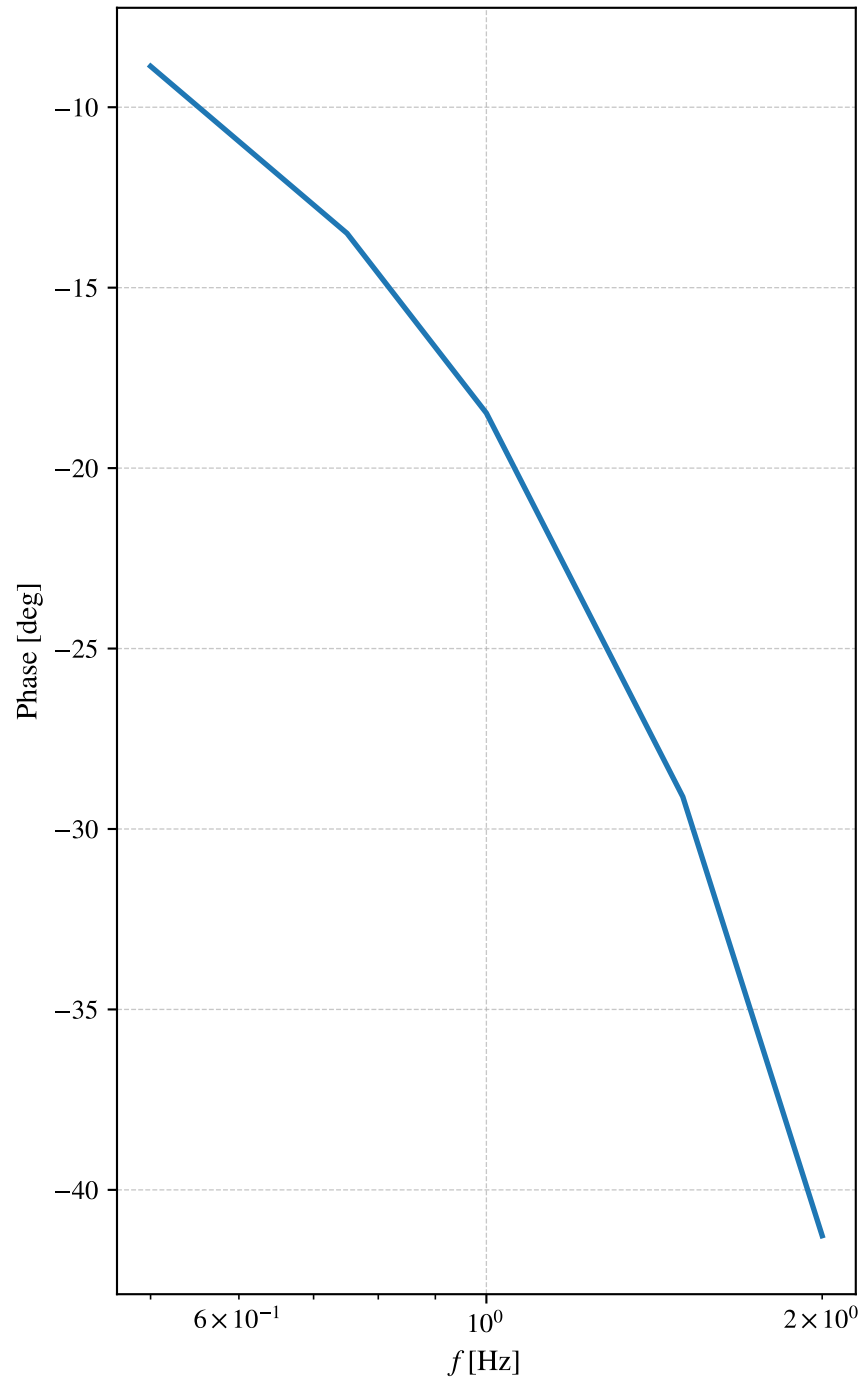


Frequency Response — Gain (ISO7401)



Frequency Response — Phase (ISO7401)

Phase: a_y vs δ_{HWA}



Phase: r vs δ_{HWA}

